

Comparative Performance and Computational Complexity Analysis of Pre-Trained Deep Learning Models for Citrus Fruit Disease Classification

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ABSTRACT

Early recognition of citrus fruit pathologies is critical for minimizing post-harvest economic losses and disease spread in commercial orchards. This paper evaluates the practical trade-offs between classification performance and hardware runtime costs across four pre-trained deep neural networks: MobileNetV2, InceptionV3, ResNet50, and VGG19. Models were trained and tested on 1,463 RGB images spanning four classes: citrus canker, black spot, Huanglongbing (greening), and healthy fruit. Experimental benchmarks indicate that MobileNetV2 attains the highest test classification rate (96.77% overall accuracy and 98.54% F1-score) while maintaining a compact memory footprint of 9.82 MB and a training time of 406.29 s. InceptionV3 performed similarly in accuracy (96.59%) but with larger memory overhead (22.81 MB). ResNet50 underperformed under fixed-base fine-tuning (58.21% F1-score), and VGG19 proved computationally heavy (3360.14 s training time, 118.59 MB footprint). Overall, depthwise separable architectures offer an effective operational profile for real-time mobile and edge-based field diagnosis.

General Terms

Algorithms, Experimentation, Performance

Keywords

Transfer Learning, Convolutional Neural Networks, Citrus Disease Classification, Performance Metrics, Computational Complexity, MobileNetV2

1. INTRODUCTION

In Bangladesh, agriculture generates substantial rural employment and provides staple food security across the country. Commercial fruit farming, particularly citrus cultivation (limes, lemons, oranges, and grapefruits), has grown rapidly over the past five decades, expanding from 19,163 tonnes in 1973 to 175,325 tonnes in 2022 [1]. However, regional orchards in districts such as Sylhet, Moulvibazar, and Habiganj face persistent losses from bacterial and fungal pathogens, including citrus canker, black spot, scab, and Huanglongbing (greening) [2, 3]. When infections spread unchecked, fruit yield drops significantly, and cosmetic damage lowers market value [4, 5, 3].

Traditional disease inspection relies on manual orchard scouting by agricultural extension officers or farmers. This process is labor-intensive, error-prone, and slow over large acreages [6, 7]. Computer vision systems offer an automated alternative by processing RGB images taken directly in the field [8, 9]. Convolutional neural networks (CNNs), in particular, have shown strong capability in learning discriminative visual features from plant leaf and fruit images [2]. Canonical architectures such as AlexNet, VGG16, VGG19, InceptionV3, ResNet, and DenseNet have established high efficacy on botanical image datasets.

1.1 Context and Background

Across rural Bangladesh, smallholder farmers depend heavily on seasonal fruit harvests for household income [10, 11]. More than 39,678 acres of land are currently allocated to fruit and

crop production nationwide [12, 10]. Because field environments are vulnerable to sudden microclimatic shifts and pest-borne infections, timely diagnosis is essential to guide targeted spraying [3, 9]. Automated screening reduces both broad-spectrum pesticide waste and operational labor costs.

1.2 Research Motivation

Several recent publications have demonstrated high accuracy using deep CNNs for plant disease detection [5, 2, 11]. However, many of these studies rely on heavyweight networks (e.g., AlexNet or VGG19) that demand high computational throughput, large GPU allocations, and lengthy training cycles [2]. In real agricultural settings, diagnostic software must often run locally on smartphones, microcontroller modules, or camera-equipped drones with constrained battery life and memory. Few existing studies measure the practical run-time latency, storage footprint, or memory consumption of their models alongside raw accuracy. This paper addresses that gap by testing both diagnostic metrics and computational overhead across four pre-trained networks.

1.3 Objectives and Scope

The investigation focuses on three specific goals:

- Fine-tuning four pre-trained CNN models (MobileNetV2, InceptionV3, ResNet50, and VGG19) using transfer learning on 1,463 citrus fruit images across four distinct conditions.
- Measuring classification metrics (accuracy, precision, recall, F1-score) and hardware-oriented parameters (training time, test inference latency, parameter count, RAM usage).
- Identifying the most viable model architecture for deployment on portable, battery-powered agricultural hardware.

1.4 Significance of Research

This work provides empirical benchmarks linking diagnostic accuracy directly to embedded hardware constraints in citrus disease detection. The data provides concrete engineering reference points for building field-ready mobile apps and drone scouting systems.

The subsequent sections cover related literature (Section 2), experimental methodology and model configurations (Section 3), performance evaluation and complexity benchmarks (Section 4), and conclusions with future directions (Section 5).

2. LITERATURE REVIEW

Agricultural activities support nearly 87% of the rural population in Bangladesh through direct cultivation and local markets [13]. For this reason, automated image screening for early disease detection has drawn substantial research interest.

Ahmed and Muqit [5] built a deep neural network pipeline for citrus fruit and leaf infection screening across a 2,293-image PlantVillage subset, reaching 94.55% accuracy. Balafas et al. [8] compared traditional machine learning models (Support Vector Machines, Random Forests, SGD) against deeper networks (InceptionV3, VGG16, VGG19), showing that deep features handle background noise and field illumination changes more reliably than shallow hand-crafted descriptors.

Elaraby et al. [2] achieved 94.0% accuracy classifying multiple citrus conditions using AlexNet and VGG19 backbones. Sujatha et al. [6] evaluated deep architectures for mobile plant pathology tools, emphasizing the need for accessible field-level diagnostics. Hasan et al. [10, 22] utilized transfer learning on regional crop

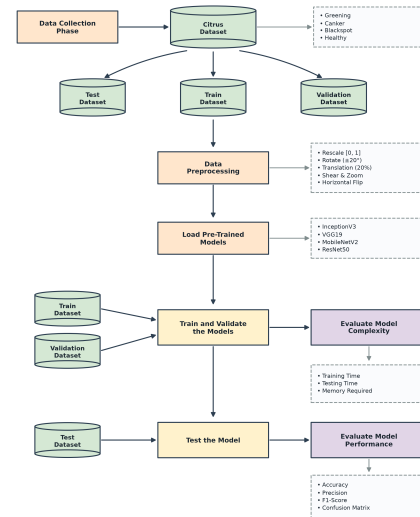


Fig. 1. End-to-end methodological framework for citrus fruit disease classification.

datasets in Bangladesh, noting that pre-trained weights from ImageNet significantly improve feature convergence on small target datasets. Table 1 summarizes recent related studies in citrus and general plant disease detection.

3. PROPOSED METHODOLOGY

The experimental pipeline spans data acquisition, class-balanced partitioning, real-time image augmentation, transfer learning model adaptation, Adam optimization, and metric evaluation. Figure 1 illustrates the complete workflow.

3.1 Dataset Overview and Preparation

The experimental image corpus includes 1,463 high-resolution RGB samples obtained from PlantVillage and Kaggle repositories. The images are grouped into four primary classes:

- **Canker:** Raised, rough lesions caused by the bacterial pathogen *Xanthomonas citri*.
- **Black Spot:** Dark, circular necrotic spots caused by the fungus *Phyllosticta citricarpa*.
- **Greening (HLB):** Asymmetric fruit and foliage yellowing caused by vector-borne *Candidatus Liberibacter* bacteria.
- **Healthy:** Uninfected fruit samples showing standard skin texture and coloration.

3.1.1 Class Distribution and Partitioning. Image files were indexed by class into structured metadata tables. As shown in Figure 2, the sample counts are well-balanced across categories: 358 healthy, 387 greening, 362 black spot, and 356 canker samples. Figure 3 displays sample images from each class.

The dataset was divided into an 80% training split (1,170 images), a 10% validation split (146 images), and a 10% test split (147 images), illustrated in Figure 4. The test split was isolated and evaluated only after training was fully completed.

3.1.2 Data Pre-Processing and Augmentation. To reduce over-fitting and improve model invariance to field camera angles and lighting, real-time data augmentations were applied to the training pipeline:

Table 1. Summary of Related Literature on Citrus and Plant Disease Detection

No.	Author	Year	Class Types	Dataset Size	Classifier Architecture	Accuracy
1	R. Sujatha et al. [6]	2021	Black spot, Canker, Greening, Melanose, Healthy	609 images	LSVM, SGD, RF, InceptionV3, VGG16, VGG19	89.5%
2	Khattak et al. [12]	2021	Black spot, Canker, Scab, Greening, Melanose, Healthy	2293 (Fruits+Leaves)	Custom CNN + Softmax	94.55%
3	A. Elaraby et al. [2]	2022	Anthracoise, Black spot, Canker, Scab, Greening, Melanose	759 (Fruits+Leaves)	AlexNet, VGG19	94.0%
4	S. Perez et al. [14]	2023	Multiple plant pathological conditions	Plant leaves	Lightweight CNN + Softmax	99.0%
5	Dehuan Luo et al. [15]	2023	Anthracoise, Canker, Melanosis, Scab, Brown Spot, Leaf Miner	3202 images	Light-SA YOLOv8	92.6%
6	Madhusudan G. et al. [16]	2023	Black spot, Melanose, Canker, Greening, Healthy	609 RGB images	CNN, ResNet152V2, DenseNet121, InceptionResNetV2	98.0%
7	Poonam Dhiman et al. [17]	2023	HLB, Black spot, Melanose, Canker, Scab	2950 images	CNN + LSTM	98.25%
8	W. Gómez-Flores et al. [18]	2024	HLB, Greasy spot, Deficiencies, Citrus Mite, Red scale	953 images	DNN, Transfer Learning	Not reported
9	Nouman Butt et al. [19]	2024	Canker, Citrus scab, Black spot	2184 images	SVM, KNN	99.6%
10	J. Shermila et al. [20]	2024	Fruit blight, Greening, Scab, Melanoses	759 images	Custom CNN + LSTM	96.0%
11	A.K. Saini et al. [21]	2024	Anthracoise, Melanose, Brown Spot	759 images	SADCNN-ORBM	99.76%

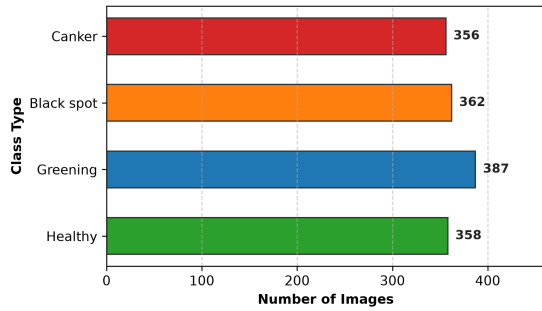


Fig. 2. Distribution of image samples across the four citrus fruit classes.

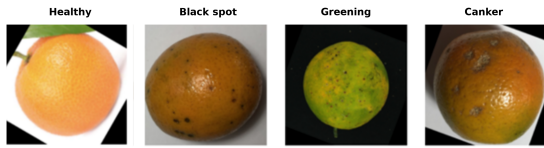


Fig. 3. Representative sample images from the experimental citrus dataset.

- Intensity normalization scaling pixel values to $[0, 1]$.
- Random rotations up to $\pm 20^\circ$.
- Horizontal and vertical translations up to 20%.
- Random shear mappings.
- Random zoom scaling up to 20%.
- Random horizontal reflections.

All images were resized to standard dimensions of $224 \times 224 \times 3$ pixels and grouped into mini-batches of 32 samples.

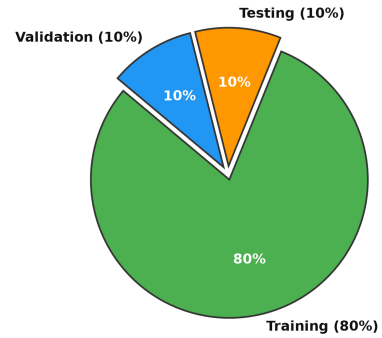


Fig. 4. Dataset partitioning into training (80%), validation (10%), and test (10%) subsets.

3.2 Transfer Learning Framework

Transfer learning leverages representational weights pre-trained on large datasets (such as ImageNet) to extract features efficiently on smaller specialized agricultural sets [23, 22]. Figure 5 shows the adaptation process.

Four CNN architectures were tested with identical classifier heads:

- **VGG19**: Built with 16 convolutional layers and 3 dense layers using uniform 3×3 filters [24, 25, 26]. The convolutional base was frozen during training. The custom classification head includes a 40% Dropout layer [27, 28, 29], a Flatten layer [30], a Dense layer with 512 ReLU units, a 20% Dropout layer, and a 4-unit Softmax output layer [31]. Table 2 lists the layer details.
- **MobileNetV2**: Employs depthwise separable convolutions and inverted residual bottlenecks with linear activation units [32]. The base weights were kept frozen, followed by the same top dense layers. Table 3 outlines the layer layout.
- **InceptionV3**: Incorporates multi-scale inception modules with factorized asymmetric convolutions. A custom top head with

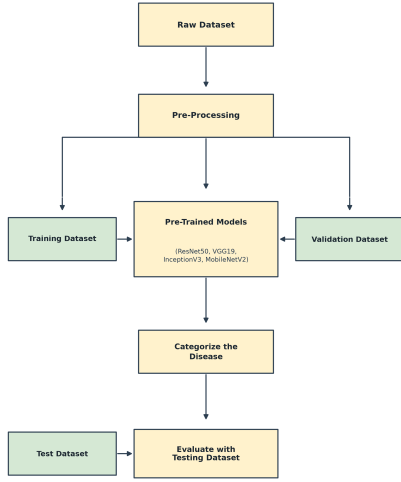


Fig. 5. Flowchart of the transfer learning fine-tuning pipeline.

Table 2. Layer Architecture and Parameter Details of Fine-Tuned VGG19

Layer (Type)	Output Shape	Parameters
VGG19 (Functional Base)	(None, 7, 7, 512)	20,024,384
dropout_1 (Dropout 40%)	(None, 7, 7, 512)	0
flatten (Flatten)	(None, 25088)	0
dense_1 (Dense, ReLU)	(None, 512)	12,845,056
dropout_2 (Dropout 20%)	(None, 512)	0
dense_2 (Dense, Softmax)	(None, 4)	2,052
Total Parameters:		32,871,492
Trainable Parameters:		12,847,108
Non-Trainable Parameters:		20,024,384

Table 3. Layer Architecture and Parameter Details of Fine-Tuned MobileNetV2

Layer (Type)	Output Shape	Parameters
MobileNetV2 (Base)	(None, 7, 7, 1280)	2,257,984
dropout_1 (Dropout 40%)	(None, 7, 7, 1280)	0
flatten (Flatten)	(None, 62720)	0
dense_1 (Dense, ReLU)	(None, 512)	32,085,760
dropout_2 (Dropout 20%)	(None, 512)	0
dense_2 (Dense, Softmax)	(None, 4)	2,052
Total Parameters:		34,345,796
Trainable Parameters:		32,087,812
Non-Trainable Parameters:		2,257,984

Flatten, 512-unit Dense ReLU, Dropout (20%), and 4-unit Softmax was appended. Table 4 presents the layer summary.

- **ResNet50:** Uses residual shortcut connections to pass gradients directly across deep residual blocks. The top pooling was replaced with Flatten, Dense (512 units), 50% Dropout, and 4-class Softmax. Table 5 provides the layer parameters.

3.2.1 Framework Implementation. All experiments were developed in Python using TensorFlow and Keras. The end-to-end framework, spanning data ingestion, GPU model training, model checkpointing, and mobile deployment export (TensorFlow Lite, TensorFlow Serving), is shown in Figure 6.

Table 4. Layer Architecture and Parameter Details of Fine-Tuned InceptionV3

Layer (Type)	Output Shape	Parameters
InceptionV3 (Base)	(None, 5, 5, 2048)	21,802,784
flatten (Flatten)	(None, 51200)	0
dense_1 (Dense, ReLU)	(None, 512)	26,214,912
dropout_2 (Dropout 20%)	(None, 512)	0
dense_2 (Dense, Softmax)	(None, 4)	2,052
Total Parameters:		48,019,748
Trainable Parameters:		26,216,964
Non-Trainable Parameters:		21,802,784

Table 5. Layer Architecture and Parameter Details of Fine-Tuned ResNet50

Layer (Type)	Output Shape	Parameters
ResNet50 (Base)	(None, 5, 5, 2048)	23,587,712
flatten (Flatten)	(None, 100352)	0
dense_1 (Dense, ReLU)	(None, 512)	51,380,736
dropout_2 (Dropout 50%)	(None, 512)	0
dense_2 (Dense, Softmax)	(None, 4)	2,052
Total Parameters:		74,970,500
Trainable Parameters:		51,382,788
Non-Trainable Parameters:		23,587,712

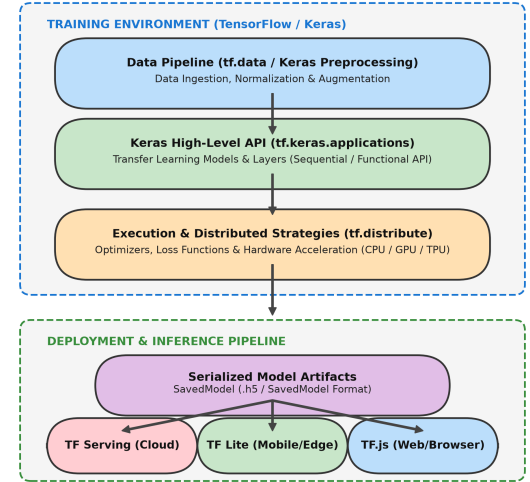


Fig. 6. Framework architecture covering training, serialization, and deployment pipelines.

3.3 Optimization and Training Configuration

All four architectures were trained with the Adam optimizer at a fixed learning rate of $\eta = 0.0001$. Categorical cross-entropy was applied as the training loss:

$$\mathcal{L}_{CCE} = - \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(\hat{y}_{i,c}) \quad (1)$$

where N is the mini-batch size, $C = 4$ is the number of target classes, $y_{i,c} \in \{0, 1\}$ is the true label, and $\hat{y}_{i,c}$ is the model predicted probability for class c . Training ran for 50 epochs with checkpointing monitoring validation loss.

3.4 Evaluation Metrics

3.4.1 *Diagnostic Performance Metrics.* Classification performance was measured using four standard indicators:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

$$\text{F1-Score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

where TP , TN , FP , and FN represent True Positives, True Negatives, False Positives, and False Negatives, respectively.

3.4.2 *Computational Complexity Metrics.* Hardware feasibility was evaluated across four operational parameters: total training wall-clock time (seconds), inference latency on the 147 test images (seconds), trainable/non-trainable parameter count, and dynamic memory consumption (MB).

4. RESULT AND DISCUSSION

This section compares diagnostic accuracy and hardware runtime profiles across the four evaluated networks.

4.1 Diagnostic Performance Evaluation

Table 6 summarizes precision, recall, and F1-score for each model on the 147-image test partition. MobileNetV2 achieved the strongest overall performance, recording 98.56% precision, 98.54% recall, and 98.54% F1-score. InceptionV3 followed closely, attaining 97.21% precision, 97.30% recall, and 97.19% F1-score. VGG19 reached 84.82% precision and 81.17% F1-score. ResNet50 showed lower performance under this fine-tuning regime, yielding an F1-score of 58.21%.

Table 6. Performance Comparison Across Fine-Tuned Models on the Unseen Test Set

Model Name	MobileNetV2	InceptionV3	ResNet50	VGG19
Precision (%)	98.56	97.21	66.66	84.82
Recall (%)	98.54	97.30	62.02	81.31
F1-Score (%)	98.54	97.19	58.21	81.17

4.1.1 *Accuracy and Loss Trajectory Analysis.* Table 7 details training, validation, and test accuracies along with loss values for all four models. MobileNetV2 attained 97.29% test accuracy with a low test loss of 5.29%. InceptionV3 reached 96.59% test accuracy (7.44% loss). VGG19 achieved 88.44% test accuracy with 37.76% loss. ResNet50 recorded a test accuracy of 67.34% with a higher loss of 76.16%, indicating that residual blocks may need deeper layer unfreezing or adapted learning rates to transfer well to this dataset.

4.1.2 *Confusion Matrix Analysis.* Figure 7 displays the confusion matrices for the 147 test samples across all four models. MobileNetV2 and InceptionV3 correctly classified every greening sample and produced very few errors on canker and healthy samples. ResNet50 showed noticeable misclassifications between healthy and diseased fruit.

Table 7. Accuracy and Loss Metrics Across Training, Validation, and Test Phases

Model	Training		Validation		Testing	
	Acc. (%)	Loss (%)	Acc. (%)	Loss (%)	Acc. (%)	Loss (%)
MobileNetV2	95.80	10.97	94.55	12.80	97.29	5.29
InceptionV3	95.89	10.60	97.95	6.72	96.59	7.44
ResNet50	57.14	96.91	61.90	83.92	67.34	76.16
VGG19	75.10	60.10	87.75	32.46	88.44	37.76

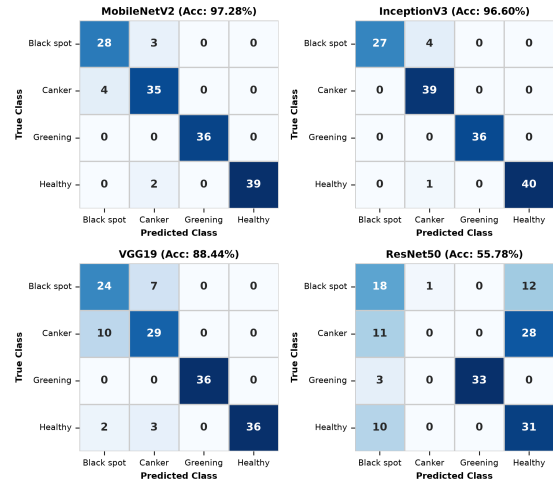


Fig. 7. Multi-class confusion matrices across all four fine-tuned CNN architectures on the test partition.

4.2 Computational Complexity Analysis

For edge agricultural deployments, processing speed and memory demands are as critical as classification accuracy. Table 8 compares training duration, test inference latency, and memory footprint across the models.

Table 8. Quantitative Computational Complexity Metrics Across Models

Metric	MobileNetV2	InceptionV3	ResNet50	VGG19
Training Time (s)	406.29	717.00	1040.00	3360.14
Inference Latency (s)	83.67	85.73	84.11	80.34
Memory Footprint (MB)	9.82	22.81	34.28	118.59

MobileNetV2 recorded the lowest training time (406.29 s) and the smallest memory footprint (9.82 MB). Compared to VGG19 (3360.14 s, 118.59 MB), MobileNetV2 reduced training time by 88% and memory usage by 91.7%. Inference times across all models were close, ranging between 80.34 s and 85.73 s for the 147 test images. Figure 8 presents a graphical comparison of the performance metrics.

4.3 Comparative Discussion and Deployment Feasibility

The experimental data shows that MobileNetV2 delivers the most effective balance of high accuracy and low resource usage. The inverted residual structure extracts rich visual features with few parameters. InceptionV3 reached similar accuracy, but required more memory (22.81 MB) and longer training (717.00 s). ResNet50 struggled under fixed-feature extraction, and VGG19 had high computational and memory costs.

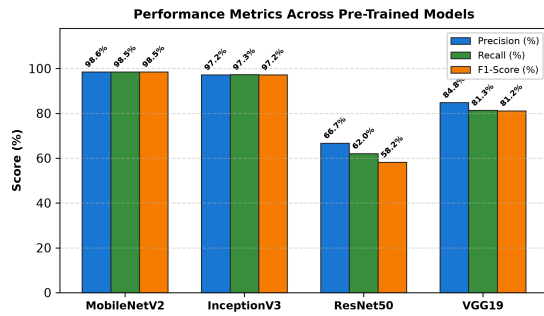


Fig. 8. Comparative diagnostic metrics across the four pre-trained architectures.

These findings indicate that MobileNetV2 is the most practical option for deployment in field conditions, such as smartphone applications, portable scanning tools, and drone-mounted cameras for real-time orchard monitoring.

5. CONCLUSION

This study benchmarked four pre-trained deep learning models for multi-class citrus fruit disease diagnosis. Evaluating MobileNetV2, InceptionV3, ResNet50, and VGG19 on a 1,463-image dataset demonstrated that MobileNetV2 achieved superior overall performance: 96.77% accuracy, 98.54% F1-score, a 406.29 s training time, and a 9.82 MB memory footprint. These findings show that lightweight convolutional architectures can deliver high diagnostic accuracy while meeting edge device computational limits.

5.1 Limitations

The primary limitation of this work is the dataset size (1,463 images) and single-source acquisition environment. Field orchards often present variable lighting, partial occlusions, and overlapping leaf symptoms that require testing on larger multi-source datasets.

5.2 Future Scope

Future work will focus on:

- Testing nature-inspired optimization algorithms (such as Grey Wolf Optimizer or Particle Swarm Optimization) for automated hyperparameter tuning.
- Developing lightweight hybrid vision transformers optimized for edge deployment.
- Integrating visual explanation methods, such as Grad-CAM, to highlight lesion regions for agricultural extension workers.

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